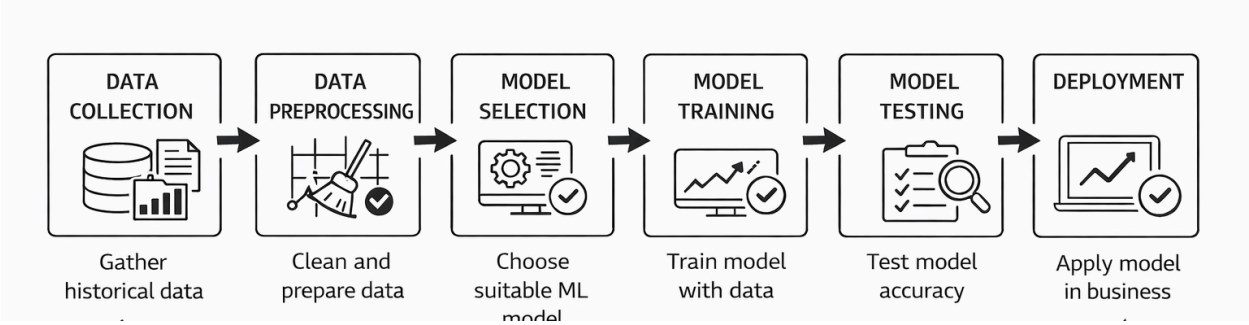
UNIT III – BUSINESS FORECASTING

MACHINE LEARNING FOR PREDICTIVE ANALYTICS

Machine Learning (ML) in predictive analytics is the use of **algorithms and data** to **predict future outcomes** such as sales, demand, or trends based on past data.

PROCESS OF ML IN PREDICTIVE ANALYTICS



STEPS INVOLVED

1. Data Collection

- Collect historical data (sales, customer data, market trends)
- Data can be structured or unstructured

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2. Data Preprocessing

- Clean data (remove errors, missing values)
- Convert data into usable format

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3. Model Selection

- Choose suitable ML model
- Example: Linear Regression, Decision Tree

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4. Model Training

- Train model using past data
- Model learns patterns and relationships

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5. Model Testing

- Test model accuracy using new data
- Check performance

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6. Prediction

- Use model to predict future outcomes
- Example: future sales, demand

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7. Deployment

- Apply model in real business environment
- Support decision making

TYPES OF MACHINE LEARNING USED

1. Supervised Learning

- Uses labeled data
- Example: Sales prediction

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2. Unsupervised Learning

- Finds patterns without labeled data
- Example: Customer segmentation

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3. Reinforcement Learning

- Learns through trial and error
- Example: Recommendation systems

COMMON ML ALGORITHMS

- Linear Regression → Predict numerical values
- Decision Tree → Classification and prediction
- Random Forest → Improved accuracy
- K-Means Clustering → Group customers
- Neural Networks → Complex predictions

APPLICATIONS IN BUSINESS FORECASTING

- Sales forecasting
- Demand prediction
- Customer behaviour prediction
- Risk analysis
- Inventory planning

ADVANTAGES

- High accuracy
- Handles large data
- Improves decision making
- Saves time and cost

2 MARKS

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1. Types of data required for ML-based forecasting models

- **Historical data** (past sales, demand)
- **Customer data** (age, behavior, preferences)
- **Market data** (trends, competition)
- **External data** (weather, economy)

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2. Define HR Planning

HR Planning is the process of **estimating the number and type of employees needed** in the future and planning how to meet those needs.

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3. Types of Sales Analytics

- Descriptive analytics
- Diagnostic analytics
- Predictive analytics
- Prescriptive analytics

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4. Why is a marketing strategy necessary?

- Helps target the right customers
- Improves sales and profit
- Gives competitive advantage
- Guides business decisions

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5. What are 7R's?

7P’s is an extended marketing mix:

- Right Product.
- RIght Quality
- Right Condition
- RIght Place
- Right Time
- Right Customer(people)
- Right Price

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6. Short notes on HR Analytics

HR Analytics is the use of **employee data to improve performance and decision-making**. It helps in recruitment, training, performance evaluation, and reducing employee turnover.

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7. Define Supply Chain Network

Supply Chain Network is the system of **suppliers, manufacturers, warehouses, and distributors** that work together to deliver products to customers.

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8. Difference: Supply vs Inventory Management

Aspect	Supply Management	Inventory Management
Meaning	Managing flow of goods from suppliers	Managing stock inside company
Focus	Procurement and supply	Storage and control of stock
Goal	Ensure supply availability	Maintain optimal inventory

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9. Four Training Steps

- Identify training needs
- Design training program
- Conduct training
- Evaluate training effectiveness

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10. How each type of sales analytics improves performance

1. **Descriptive** → Shows past sales performance
2. **Diagnostic** → Finds reasons for success/failure
3. **Predictive** → Forecasts future sales
4. **Prescriptive** → Suggests actions to improve sales

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11. How marketing strategy helps achieve goals?

Marketing strategy helps by **targeting the right customers, improving product positioning, increasing sales, and achieving business objectives effectively**.

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12. Common software for sales analysis

- Microsoft Excel
- Power BI
- Tableau
- Salesforce
- Google Analytics

13. Define Data Mining.

Data Mining is the process of **extracting useful patterns and knowledge from large datasets** using techniques like clustering and classification.

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14. What is a Predictive Analytics Model?

A Predictive Analytics Model is a method that uses **historical data and algorithms to predict future outcomes** like sales or demand.

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15. List the types of Data Mining techniques.

- Classification
- Clustering
- Association rule mining
- Regression

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16. Differentiate Data Mining and Machine Learning.

Data Mining	Machine Learning
Extracts patterns from data	Learns patterns to predict outcomes
Used for analysis	Used for prediction

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17. Applications of Machine Learning in business forecasting

- Sales prediction
- Demand forecasting
- Customer behaviour prediction
- Risk analysis

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18. Process of Data Mining

- Data collection
- Data cleaning
- Pattern discovery
- Evaluation and interpretation

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19. Define Recruitment in HR.

Recruitment is the process of **finding and selecting suitable candidates** for a job.

Steps: Identify vacancy → Source candidates → Screening → Interview → Selection → Hiring.

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20. Differentiate Training and Development

Training	Development
Short-term	Long-term
Focus on specific skills	Focus on overall growth

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21. Applications of HR Analytics

- Employee performance analysis
- Recruitment improvement
- Training effectiveness
- Attrition prediction

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22. What is Demand Planning?

Demand Planning is the process of **forecasting customer demand** to ensure proper production and supply.

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23. Differentiate Logistics and Supply Chain

Logistics	Supply Chain
Focus on transportation	Covers entire process
Limited scope	Wide scope

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24. Applications of Marketing Analytics

- Customer segmentation
- Campaign analysis
- Sales prediction
- Market trend analysis

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25. Process of Customer Behaviour

- Problem recognition
- Information search
- Evaluation of alternatives
- Purchase decision
- Post-purchase behaviour

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26. Marketing Strategies

Marketing strategy is a **plan to reach target customers and increase sales**. Includes **segmentation, targeting, positioning, and promotion methods**.

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27. 4Ps (Marketing Mix)

4Ps are the key elements of marketing:

1. Product → What is sold
2. Price → Cost of product
3. Place → Where it is sold
4. Promotion → How it is advertised